

CLAUDE CODE PROMPT — “Now” Improvements for Summer Math Battle Tutor

Copy everything below this line into Claude Code, run from the repo root of `juliarus768-cyber/summer-math-battle-tutor`.

Context — read before writing any code

You are working on a gamified math learning app used daily by two real children. Project documentation lives in the repo root (or `/docs`). **Read these files first and follow them as binding rules:**

1. `PROJECT.md` — vision and non-negotiable ground rules
2. `GAME_DESIGN.md` — the exact economy, streak, and reward behavior to preserve
3. `EDUCATION.md` — the mandatory wrong-answer teaching flow you will implement
4. `CHARACTERS.md` — child profiles, personalization and privacy rules
5. `DEVELOPMENT_RULES.md` — architecture, workflow, testing, code map
6. `FUTURE_IDEAS.md` — the “Now” bucket, which this prompt implements

Key architectural facts (verify against `DEVELOPMENT_RULES.md`):

- The entire app is `index.html` (~4,700 lines: inline CSS + markup + one inline script). Navigate by the `/* ==== SECTION ==== */` banners, not line numbers. `src/App.jsx` is a dead, non-mounting prototype — do NOT edit it, do NOT use it as a behavior reference.
- State = one `localStorage` key `smbt-state-v2` via the `Store` wrapper. Any new state fields MUST be back-filled for existing saves in the existing back-fill loop after `FRESH`. Never break or erase an existing save.
- Vite build, Vercel auto-deploys `main`. Bump `APP_VERSION` when done.

Hard constraints (violating any of these fails the task)

- **Zero visual or behavioral regression** outside the changes specified here. Do not

restyle, rename, or “clean up” anything not listed.

- Wrong answers must NEVER reduce XP, coins, or streaks.
- No punishment language anywhere. Feedback praises effort, persistence, careful thinking, and improvement.
- Never compare the two children against each other.
- `main` stays stable: all work happens on feature branches with PRs. Do not merge without Yulia’s review.
- Existing child progress must survive every change (test with a pre-existing `smbt-state-v2` blob, not just a fresh state).
- Tablet-first: manually verify every touched screen at ~810×1080 (iPad) and a ~1200px Android tablet width, plus one phone width. No horizontal scroll, no hover-only interactions, touch targets stay large.

Work plan — 4 branches, 4 PRs, in this order

PR 1 — branch `fix/content-and-economy-cleanup` (small, do first)

Task 1.1 — Remove Leila and Emily everywhere. They are no longer Katya’s friends and must never appear. Search the entire `index.html` (and any content files) for `Leila` and `Emily` — check the `FRIENDS` array in the question engine, Katya’s mission story templates, word problems, and any generated-text pools. Replace with existing approved names only: `Aryna` (Katya’s friend, lives in Alberta), family members per `CHARACTERS.md` (Grandma Olga, Grandpa Serezha, Tolik the Elf, etc.), or neutral characters. Keep story quality equal — don’t just delete lines and leave grammar broken.

Task 1.2 — Fix the XP level-up inconsistency. `markMissionDone` (parent dashboard) levels up with `xpMax × 1.15` while the rest of the app uses `xpMax += 50`. Unify to `+50`. Best: extract one shared `gainXP(child, amount)` / level-up helper and call it from every place XP is awarded, so the formula can never diverge again.

Task 1.3 — Verify default PIN handling. Do not change the PIN in code. Instead: confirm the parent dashboard’s change-PIN flow works, and add a small, dismissible one-time notice on the parent dashboard if the PIN is still `1234` (“Change your PIN in Settings”).

Acceptance: `grep` for `Leila|Emily` returns nothing; leveling math is identical from every XP source; existing saves load unchanged.

PR 2 — branch `feature/teaching-loop` (the most important change)

Implement the mandatory wrong-answer teaching loop from EDUCATION.md in the **Daily Mission** flow (then reuse for Guided Mission if straightforward; Battle Arena keeps its faster 3-tier hint behavior for now).

Required behavior per question:

1. Up to **3 attempts**. Attempt counter visible but gentle (e.g., three dots).
2. After wrong attempt 1: automatically surface Brain Boost tier 1 (clue) with an encouraging line ("Not yet — here's a clue"). Never a bare "Incorrect."
3. After wrong attempt 2: automatically surface a stronger tier (Winning Strategy + Similar Example from the existing `STRATEGY_LIBRARY`).
4. After wrong attempt 3: show a **step-by-step explanation of this exact question** ending with the correct answer. Reuse the topic's strategy content, but the walkthrough must use the live question's numbers. Then mark the question as "learned, not yet earned" (no XP penalty — simply not counted correct).
5. **Reinforcement:** queue 1–3 similar questions of the same topic and difficulty (use the existing generators) to appear later in the same mission (extend the mission length by the reinforcement count, cap total at ~26 so sessions stay in the 15–20 minute band). Answering a reinforcement question correctly awards normal per-question XP plus a small "comeback" bonus (+4 XP) with a persistence-praising message.
6. **Recurring-mistake tracking:** record misses per topic in mastery data (fields likely exist — extend if needed, with save back-fill). Topics with repeated misses already get weight 4 in the adaptive engine; verify the new reinforcement data feeds that weighting.
7. XP, coins, and streaks are never reduced by any of this.

UI: build inside the existing mission screen and Brain Boost panel styling — same neon-arcade look, same fonts, same components. No new visual language.

Acceptance: a scripted wrong-wrong-wrong sequence shows escalating help, then a worked solution, then reinforcement questions later in the run; state persists across refresh mid-mission or fails safely without data loss; existing saves unaffected.

PR 3 — branch `feature/grade-age-profiles`

Task 3.1 — Birthday-driven age + advancing grade. Add to each child's profile state (with back-fill): `birthday` (Alex 2013-12-29 , Katya 2016-05-04), `currentGrade` (Alex 7, Katya 4 until Sept 2026), `gradeAdvanceMonth`: 9 . Compute age from birthday. Auto-advance `currentGrade` each September 1 (Toronto time) unless the parent has set a manual override; add a simple grade editor in the parent dashboard. All "GRADE 7→8 / GRADE 4→5"

labels must derive from `currentGrade` and `currentGrade + 1` instead of hardcoded strings. The question engine's grade-band selection should read `currentGrade` too. Frame the current summer content as the optional "Summer Bridge" mode label, per PROJECT.md.

Task 3.2 — Personal details become data. Create a `PERSONALIZATION` data object (per child) holding: display name, nickname (Alex/Sasha), theme, interests list, friend names, family member names/roles, favorite foods/rewards vocabulary — populated from CHARACTERS.md. Refactor story templates and generators to pull names/interests from this object instead of scattered string literals. Do not change any visible story output except where Task 1.1 already did. This prepares parent-editable profiles later; no editing UI needed yet.

Acceptance: changing `currentGrade` in the parent dashboard updates labels and question banding; no hardcoded `Leila` -style literals remain in generators; existing saves back-fill cleanly.

PR 4 — branch `test/playwright-baseline`

Set up Playwright properly (config, `npm test` script, CI-friendly headless run against `vite preview`). Write the baseline suite from DEVELOPMENT_RULES.md → Testing:

- app loads; profile picker renders both children + parent
- selecting Alex/Katya lands on their themed home
- Daily Mission starts; 20 questions display; numeric answers enterable via touch-sized inputs (test a tablet viewport: 810×1080, and a phone viewport)
- correct answer → progress + XP; wrong answers → the PR 2 teaching flow (3 attempts, escalating hints, worked solution, reinforcement queued)
- mission completion updates XP, coins, streak; values persist after reload
- store: request a reward → pending state; parent PIN gate blocks without PIN; approve deducts coins; decline deducts nothing
- navigation: Switch always returns to profile picker; no dead ends
- refresh mid-app does not destroy progress

Seed tests with a fixture `smbt-state-v2` blob so they're deterministic. Keep the suite fast (<2 min).

Acceptance: `npm test` passes headless; README gains a short "Running tests" section.

For every PR

Follow the pre-merge checklist in DEVELOPMENT_RULES.md: run tests, manual tablet check, no visual regression, existing data preserved, production build succeeds. Write a clear PR description listing behavior changes and how you verified data safety. Bump `APP_VERSION` to `1.2.0` in the final PR and note the changes in the in-app update notes if that mechanism exists.

Out of scope — do not do

- No React/TypeScript migration, no build tooling changes, no restyling.
- Do not touch `src/` except leaving it as-is.
- Do not add a backend, accounts, or any external API.
- Do not add new subjects, modes, or rewards.
- Do not commit any real personal data to logs, test snapshots, or fixtures beyond what already exists in the app (CHARACTERS.md privacy rules).

Finally: report back anything you discover that contradicts the documentation (wrong constant, missing function, undocumented behavior) so the docs can be corrected — recommend exactly which doc the correction belongs in.